

## Proposal 1

### MPhil/PhD Proposal

#### *Neurocognitive correlates of abuse and neglect:*

#### *Genotype driven differences in baseline functioning and response to therapeutic intervention*

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#### ***Background:***

Children who experience early adversity are more likely to suffer from a range of psychosocial and neurocognitive difficulties compared to typically developing (TD) children. More specifically, exposure to early maltreatment (MT) and neglect has been shown to increase risk for adult psychopathology as well as antisocial behaviour later in life, imposing a significant financial burden on both government and mental health services (Kaplow & Widom, 2007). For example, research has demonstrated that persistent early maltreatment is correlated with the development of a number of mental illnesses – including depression, post-traumatic stress disorder, and anxiety – as well as behavioural problems, such as substance abuse and offending behaviour (Oswald, Heil, & Goldbeck, 2009). Neuropsychological studies looking at TD and MT children have found decreased cognitive ability and executive functioning, as well as abnormalities in processing of negative emotions, in children with MT (See e.g. DePrince, Weinzierl, & Combs, 2009; as well as McCrory, DeBrito & Viding, *in press* for a review). Brain imaging studies have found both structural and functional differences between MT and TD children, whereby MT is particularly seen to affect the development of brain regions associated with emotion processing and regulation, as well as higher order cognitive functioning (McCrory, DeBrito, & Viding, *in press*). Neuroendocrine studies have highlighted how MT is associated with dysregulation of the Hypothalamic-Pituitary-Adrenal (HPA) [axis, which in turn has been associated with increased risk of psychopathology](#) (McCrory, DeBrito, & Viding, *in press*). More recently, genetic studies have suggested that specific polymorphisms in genes associated with stress reactivity may moderate the likelihood of an individual developing psychopathology following MT (Caspi & Moffitt, 2006). Therefore, overall, there is convincing evidence from a wide array of disciplines that MT influences an individual's development at biological, cognitive and behavioural levels.

Despite this general consensus there is a dearth of research providing a comprehensive neuropsychological assessment in children with MT. Furthermore, neurocognitive functioning in MT children has not been related to broader aspects of functioning (e.g. behaviour and family factors). In addition, there is little data charting whether interventions are associated with changes in neurocognitive functioning, particularly in adolescents with MT. Finally, both the baseline neurocognitive functioning and the effect of intervention may be moderated by differences in genetic make-up in children with MT. This PhD will address each of these areas of research by recruiting a large sample of MT adolescents in collaboration with Kids-Company, a charity that has an

unprecedented record of successful intervention with MT adolescents. Details of the planned studies are outlined below.

***Proposed PhD research:***

***Study 1: Baseline neurocognitive profile of MT adolescents***

***Study 2: Neurocognitive profile pre- and post-intervention***

***Objectives:***

Study 1: To obtain an overall index profile of neurocognitive function amongst MT adolescents compared to TD adolescents and relate this profile to measures of psychosocial functioning and behaviour.

Study 2: Explore whether intervention affects the baseline neurocognitive profile by assessing overall neurocognitive function prior to and after intervention.

***Participants:*** 200 MT adolescents will be recruited in schools in collaboration Kids-Company. The comparison group will consist of 50 TD adolescents, matched for age, socio-economic status, and cognitive ability, recruited from the same schools as MT adolescents.

***Method:*** An accelerated longitudinal design will be adopted. Out of the 200 MT group, 100 will have recently entered service provision in Kids-Company, while the remaining 100 will have had regular Kids-Company services for 3 years. This method will maximise the chance of detecting intervention effects on neurocognitive functioning. All participants (MT and TD) will be assessed both at the start of the PhD and after 12-18 months post first assessment. The measures detailed below will be rated for the overall group of participants by parents or carers as well as teachers. Those measures that have a self-rating version will be rated by participants themselves. For the MT group only, measures will also be rated by key-workers.

***Measures of Neurocognitive Function:*** In order to assess neurocognitive function a battery of tasks will be used, including: 1) the *NEPSY* neuropsychological tests series (Korkman, Kirk, & Kemp, 1998) that chart executive function, communication and social perception; 2) the *intradimensional/extradimensional set shift task* (ID-ED; Owen et al., 1991) is an additional executive function task tapping into processes associated with orbitofrontal and dorsolateral prefrontal cortex functioning respectively; 3) a task of effortful emotion regulation (Ochsner et al., 2004) that engages prefrontal and anterior cingulate regions implicated in cognitive control; 4) and an *emotional face recognition task* that taps into amygdala functioning.

***Measures of Child Characteristics:*** A range of established tasks and questionnaires will be used to obtain an overall profile of child characteristics. These will include the *Strengths and Difficulties Questionnaire* (Goodman, 1997); *Adolescent Symptom Inventories* (ASI; 4<sup>th</sup> Ed., Gadow & Sprafkin, 2002); the *Inventory of Callous-Unemotional Traits* (ICU; Frick, 2003); a self report measure of delinquency (Huizinga & Elliott, 1986); the Basic Empathy Scale (Jolliffe & Farrington, 2006) and the *Wechsler Abbreviated Scale of Intelligence* (WASI; [Wechsler, 1999](#)) as a measure of IQ.

***Measures of Psychosocial Functioning:*** Psychosocial functioning will be assessed via a number of standardised questionnaires indexing perceptions of violence and bullying, family and peer relationships. An adapted version

of the Kaufman measure (Kaufman et al., 1994) will be used along with data obtained from Kids-Company in order to characterise the risk and protective factors for adolescents in the MT group. The Friendship Quality Questionnaire (Parker & Asher, 1993) will be administered to both TD and MT groups as a self-report measure of the participant's perceived quality of relationships with peers. Outcome values measure (Pardini et al., 2003) will be used to assess perceptions of violence and bullying in response to short vignettes. Additionally, Kids-Company service provision will be measured by recording factors such as frequency of meetings with key workers and number of therapy sessions attended.

***Predictions:***

Study 1: MT adolescents will show difficulties in tasks related to executive functioning, emotion regulation and basic emotion processing. An exploratory aim will be to delineate the relative strengths as well as the weaknesses of the MT sample and relate these to resilient functioning.

Study 2: Improvement in a number of neurocognitive domains is predicted post-intervention. Specifically, an improvement in emotion recognition and regulation is predicted to occur. It is also predicted that these changes will be related to improvements in the adolescent's psychosocial functioning.

***Study 3: Role of genotype in moderating baseline and intervention neurocognitive profile***

***Objective:*** To identify a) whether the baseline neurocognitive profile differs as a function of specific polymorphisms in genes associated with stress reactivity and b) whether 'stress system polymorphisms' moderate the effect of intervention on neurocognitive functioning.

***Participants:*** As above.

***Method:*** Approximately 12-18 months following commencement of PhD, adolescents from the MT group of children will be invited to provide cheek swabs for DNA. In addition to studying the effects of specific genes - by dividing the MT group to risk vs. protective allele carriers - it is also possible to generate a composite genotype risk index by quantifying the total number of risk alleles per individual. This score can in turn be related to neurocognitive measures pre- and post intervention.

***Measures:*** As above. Examples of candidate genes include monoamine oxidase-A (MAOA), serotonin transporter (5-HTT), catechol-o-methyl transferase (COMT), dopamine transporter (DAT1), and genes associated with HPA axis functioning, such as Corticotropin-releasing hormone polymorphism (CRH). This list will be updated to reflect current findings in psychiatric genetics.

***Predictions:*** Genotype will be found to moderate baseline neurocognitive profile as well as the effect of intervention.

***Selected Key References:***

- McCrory, E., DeBrito, S., & Viding, E. (*in press*). The neurobiology and genetics of maltreatment and adversity. *Journal of Child Psychology and Psychiatry*.
- Caspi, A., & Moffitt, T.E. (2006). Gene-environment interactions in psychiatry: joining forces with neuroscience. *Nature Reviews. Neuroscience*, 7, 583-590.